SEXPERIMENT NO. 2

Classification using SVM and Random Forest

1. Aim

To implement, train, and evaluate classification models using Support Vector Machine (SVM) and Random Forest classifiers on the Iris dataset, comparing their performances via accuracy score and confusion matrices.

2. Procedure

1. Import the required Python libraries, including scikit-learn, seaborn, and matplotlib.
2. Load the Iris dataset using load_iris() and separate the features (X) and target labels (y).
3. Scale the feature values using StandardScaler.
4. Split the scaled dataset into training (70%) and testing (30%) sets using train_test_split with random_state=42.
5. Create and train an SVM classifier with a linear kernel, predict the test labels, and calculate its accuracy and confusion matrix.
6. Create and train a Random Forest classifier with 100 estimators and random_state=42, then calculate its accuracy and confusion matrix.
7. Display heatmaps of both confusion matrices using seaborn and matplotlib.

3. Code

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# 2. Feature scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 3. Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.3, random_state=42
)

# Part A: Support Vector Machine (SVM)
svm_model = SVC(kernel='linear')  # You can also try 'rbf', 'poly'
svm_model.fit(X_train, y_train)
y_pred_svm = svm_model.predict(X_test)

print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
print("SVM Confusion Matrix:\n", confusion_matrix(y_test, y_pred_svm))
```

```
# Part B: Random Forest
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

# Prediction
y_pred_rf = rf_model.predict(X_test)

print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Random Forest Confusion Matrix:\n", confusion_matrix(y_test, y_pred_rf))

# Visualization
plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
sns.heatmap(
    confusion_matrix(y_test, y_pred_svm), annot=True, cmap='Blues', fmt='d'
)
plt.title("SVM Confusion Matrix")

plt.subplot(1, 2, 2)
sns.heatmap(
    confusion_matrix(y_test, y_pred_rf), annot=True, cmap='Greens', fmt='d'
)
plt.title("Random Forest Confusion Matrix")

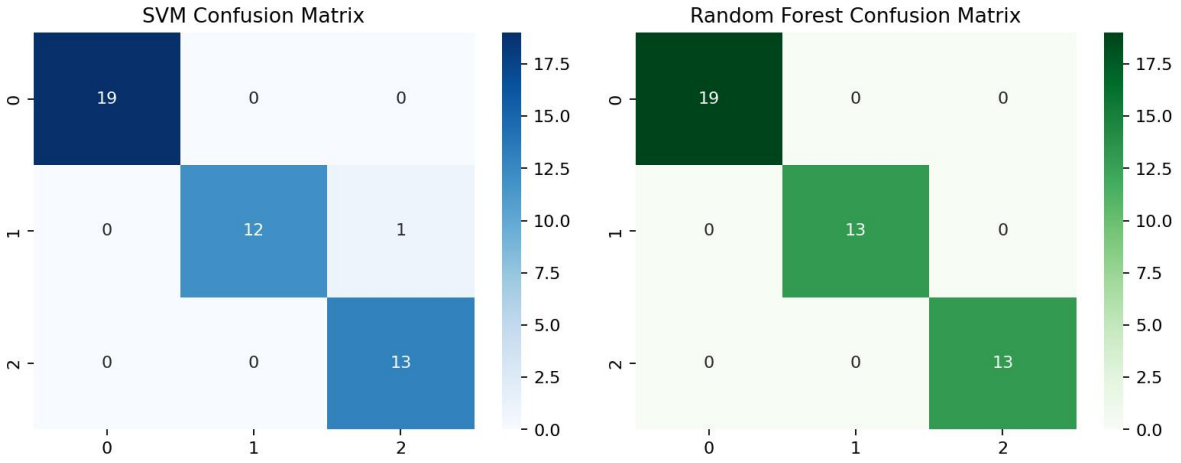
plt.tight_layout()
plt.show()
```

4. Output

Exact console output from running the code:

```
SVM Accuracy: 0.9777777777777777
SVM Confusion Matrix:
 [[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
Random Forest Accuracy: 1.0
Random Forest Confusion Matrix:
 [[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

Confusion matrix heatmaps:



5. Result

The classification algorithms were successfully implemented on the Iris dataset. The Support Vector Machine model achieved an accuracy of 97.78%, while the Random Forest classifier achieved 100.00% accuracy on the test set. Both models were evaluated using confusion matrices and visualized using heatmaps.